

# Draft Night

## V1 Product & Technical Specification

Private fantasy-football draft management application • Version 1.0 • August 2026

### 1. Product Summary

Draft Night is a private, league-specific web application for running a live fantasy-football snake draft. It provides a commissioner console, mobile-friendly team interfaces, and a read-only TV/display view. V1 supports keepers and pre-draft draft-pick trades. Draft-pick trades are represented by editing the owner of individual draft slots before the draft begins; there is no live trading workflow.

### 2. V1 Goals

- Make draft setup fast and reliable.
- Represent keeper selections cleanly.
- Allow individual draft slots to be reassigned to another team to reflect pre-draft traded picks.
- Provide a simple, mobile-friendly interface for team managers.
- Provide a strong commissioner interface with override and undo capabilities.
- Keep all connected clients synchronized in real time.
- Provide a polished TV/projector view for the room.
- Preserve the completed draft and allow results to be exported.

### 3. Explicitly Out of Scope for V1

- Auction drafts and bidding
- Draft timers
- Live draft-pick trading
- Player trades
- Waivers/free agency
- Weekly fantasy scoring or matchup management
- ESPN/Yahoo/Sleeper synchronization
- Chat/social features
- Payments or public league discovery
- AI draft recommendations or advanced projections
- Full user-account/password system

### 4. Core Design Principle

The server-side draft state is the single source of truth. UI clients submit actions; the draft engine validates and commits them; the resulting authoritative state is then broadcast to all connected clients.

The live draft should operate on an explicit sequence of draft slots. This makes pre-draft pick trades simple: the commissioner edits the drafting team assigned to a particular slot rather than recording a trade transaction.

## 5. User Roles

## 6. Draft Configuration Model

The draft has three major configuration inputs: teams, keepers, and the draft-slot grid.

### 6.1 Draft Slot

Each slot should store both its original owner and its current drafting owner.

Example: if Team A traded its Round 5 pick to Team B before the season, the Round 5 slot simply has Original Team = A and Drafting Team = B.

### 6.2 Generate Then Edit

- Generate the standard snake order automatically.
- Display the complete draft grid.
- Allow the commissioner to edit individual cells/dropdowns.
- Use the edited grid as the authoritative draft sequence.
- Validate the resulting grid before the draft can start.

## 7. Snake Draft Logic

For N teams, the normal order alternates direction by round.

The generated order is only the starting point. The explicit draft-slot grid becomes authoritative after the commissioner edits any slots for traded picks.

## 8. Keepers

Keepers are configured before the draft. A keeper consists of a team, player, and keeper round. The keeper occupies that team's draft slot in the specified round and is permanently unavailable to other teams.

V1 should deliberately support this simple keeper model rather than attempting to encode every possible league-specific keeper rule.

## 9. Player & Ranking Data

Players and rankings should be separate concepts. V1 should support CSV import.

Stable player IDs are preferred. Name matching can be supported as a fallback for manually prepared CSVs.

## 10. Live Draft Workflow

- Find the next unfilled draft slot.
- Read that slot's drafting team.
- Display that team as being on the clock.
- Team manager or commissioner submits a player selection.
- Server validates the slot, team, and player.
- Record the pick transactionally.
- Mark the player unavailable.
- Advance to the next unfilled slot.
- Broadcast the new authoritative state to all clients.

## 11. Commissioner Console

The commissioner interface is the primary administrative surface.

- League/setup configuration
- Draft order and individual slot editing
- Keeper configuration
- Player/ranking import
- Current pick display
- Player search/filter/sort
- Make pick for any team
- Undo most recent pick
- View draft history
- View team rosters
- Review validation warnings
- Start/reopen draft
- Export results

## 12. Team Manager Interface

The team view should be optimized for phones around 390×844 while remaining usable on desktop.

- Clearly show whether the team is on the clock.
- Show current round and pick.
- Search players.
- Filter by position.
- Sort by ranking/ADP/name.
- Show player rank and relevant metadata.
- Show own roster.
- Show next upcoming pick.
- Submit a pick when authorized.
- Show draft history and current board.

## 13. TV / Display View

The display is read-only and designed for a TV/projector.

- Current team and pick number
- Round
- Recent picks
- Draft board
- Team rosters or roster summaries
- Keeper indicators
- Optional top available players

- Optional indication that a slot originated with a different team

## 14. Draft Board

Recommended visual model: rows represent rounds and columns represent teams/slots.

The board must reflect the current drafting owner of each slot. It should not imply that the original owner made a pick after a pre-draft pick transfer.

## 15. Draft State Lifecycle

A commissioner should be able to reopen a completed draft if a correction is necessary.

## 16. Validation Before Start

- All teams configured.
- Expected number of draft slots generated.
- Every draft slot has exactly one drafting team.
- Keeper players are unique.
- Keeper rounds are valid.
- Keeper selections do not conflict.
- Draft slot distribution is reviewed for obvious errors.
- No unavailable/duplicate player conflicts exist.
- Draft can be locked and started.

Useful warning: flag teams that own substantially more or fewer draft slots than expected so accidental edits are caught before draft night.

## 17. Database Schema

draft\_events is optional but recommended for audit/debugging. A separate draft-pick-trade table is not required for V1.

## 18. Real-Time Architecture

Use WebSockets or Socket.IO for live synchronization.

Recommended flow:

Client → server action → validation → database transaction → authoritative state → broadcast → all clients update.

Clients should not assume a pick succeeded until the server confirms it.

## 19. Concurrency & Reliability

- Only one successful pick may occupy a draft slot.
- A player may only be selected once.
- Server-side validation is mandatory.
- Pick writes should be transactional.
- Commissioner overrides must go through the same authoritative draft engine.
- Disconnected clients should reconnect and fetch current state.
- A browser refresh should not lose draft state.

## 20. Access Model

V1 does not require a traditional account/password system. Use league/team access tokens or shareable URLs.

Tokens should be cryptographically random and database IDs should not be exposed as access credentials.

## 21. Recommended Technology

There is no requirement to use Plotly Dash. The application is better treated as a conventional interactive web app. The technology choice should follow the developer's strengths rather than the earlier dashboard-oriented stack.

## 22. Suggested Project Structure

```
app/  
  frontend/  
  backend/  
  draft/  
  engine  
  validation  
  state  
  models/  
  services/  
  api/  
  database/  
  tests/
```

## 23. Draft Engine API

The core business logic should be independent of the UI.

- generate\_draft\_slots()
- update\_draft\_slot\_owner(slot\_id, team\_id)
- add\_keeper(team\_id, player\_id, round)
- remove\_keeper(keeper\_id)
- validate\_draft\_configuration()
- start\_draft()
- current\_slot()
- available\_players()
- make\_pick(slot\_id, team\_id, player\_id)
- undo\_last\_pick()
- team\_roster(team\_id)
- draft\_history()
- export\_results()

## 24. Testing Strategy

Prioritize draft-engine tests over UI tests.

- Standard snake-order generation
- Edited draft-slot ownership
- Keeper slot behavior
- Player uniqueness
- Pick uniqueness
- Undo behavior
- Commissioner override
- Draft completion
- Concurrent pick attempts
- Reconnect/reload behavior
- Validation of malformed draft configurations

## 25. V1 Acceptance Test

A realistic end-to-end test should look like this:

- Create a 12-team, 15-round league.
- Generate snake draft order.
- Edit several individual slots to reflect pre-draft traded picks.
- Assign keepers with keeper rounds.
- Validate and lock the draft.
- Open commissioner, team, and TV clients simultaneously.
- Make picks from team devices.
- Confirm every client updates in real time.
- Reach a traded slot and verify the edited drafting team is on the clock.
- Reach a keeper slot and verify the keeper is represented correctly.
- Attempt to draft an unavailable player and verify rejection.
- Use commissioner override.
- Undo a pick and verify exact restoration of state.
- Complete the draft.
- Export final draft and roster results.

## 26. Development Plan

## 27. Definition of Done

V1 is done when a real league can configure its teams, keepers, rankings, and pre-draft traded picks; lock the configuration; run the entire snake draft from multiple phones; display the draft on a TV; recover from ordinary mistakes; and export the final results without needing a spreadsheet or manual bookkeeping.

## 28. Product Direction

Draft Night should remain intentionally narrow: a beautiful, reliable, league-specific draft-night command center. The defining capabilities are keeper support, explicit draft-slot ownership, a strong commissioner console, mobile team drafting, and a shared

TV display. Avoid expanding into a general fantasy-football platform unless a future need clearly justifies it.

End of V1 Specification

## Appendix — Tables from the Specification

| Role         | Primary capabilities                                                                                                                                       |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Commissioner | Configure league; edit draft slots; configure keepers; import players/rankings; start draft; make picks for any team; undo; correct state; export results. |
| Team Manager | View draft; search/filter players; view roster and future picks; make a pick when their team is on the clock.                                              |
| TV / Display | Read-only live view of current pick, recent picks, draft board, rosters, and optional top available players.                                               |

| Field         | Meaning                                        |
|---------------|------------------------------------------------|
| Round         | Draft round number                             |
| Pick number   | Overall pick number or round-relative position |
| Original team | Team assigned by the normal draft order        |
| Drafting team | Team currently entitled to make the selection  |
| Status        | Open, filled, or keeper-occupied               |

| Round | Normal order example (4 teams) |
|-------|--------------------------------|
| 1     | A → B → C → D                  |
| 2     | D → C → B → A                  |
| 3     | A → B → C → D                  |
| 4     | D → C → B → A                  |

| Field  | Purpose                            |
|--------|------------------------------------|
| Team   | Team retaining the player          |
| Player | Player being retained              |
| Round  | Draft round consumed by the keeper |

| Player fields | Ranking fields                   |
|---------------|----------------------------------|
| player_id     | player_id                        |
| name          | rank                             |
| position      | ADP (optional)                   |
| NFL team      | ranking source (optional)        |
| status        | league-specific notes (optional) |

| State | Behavior                                                         |
|-------|------------------------------------------------------------------|
| SETUP | Teams, players, rankings, keepers, and draft slots are editable. |
| READY | Configuration passes validation and is ready to start.           |



| State     | Behavior                                                      |
|-----------|---------------------------------------------------------------|
| LIVE      | Draft picks can be made; configuration is effectively locked. |
| COMPLETED | All required selections are complete; results are preserved.  |

| Table        | Core fields                                                                    |
|--------------|--------------------------------------------------------------------------------|
| leagues      | id, name, season, num_teams, num_rounds, status                                |
| teams        | id, league_id, name, draft_position, manager_name, access_token                |
| players      | id, name, position, nfl_team, status                                           |
| rankings     | id, league_id, player_id, rank, adp                                            |
| draft_slots  | id, league_id/draft_id, round, pick_number, original_team_id, drafting_team_id |
| keepers      | id, draft_id, team_id, player_id, round                                        |
| picks        | id, draft_id, draft_slot_id, team_id, player_id, pick_type, timestamp          |
| draft_events | id, draft_id, event_type, payload, timestamp                                   |

| Access       | Example                 |
|--------------|-------------------------|
| Commissioner | /draft/ABC123/admin     |
| Team         | /draft/ABC123/team/X82K |
| TV           | /draft/ABC123/display   |

| Layer      | Recommendation                                                  |
|------------|-----------------------------------------------------------------|
| Frontend   | React / Next.js + TypeScript, or another conventional web stack |
| Backend    | FastAPI/Python or Node/TypeScript                               |
| Database   | PostgreSQL                                                      |
| Real-time  | WebSockets / Socket.IO                                          |
| Styling    | Tailwind or preferred CSS system                                |
| Deployment | Docker + simple cloud host                                      |

| Phase | Deliverable                             |
|-------|-----------------------------------------|
| 1     | Draft engine + database + tests         |
| 2     | Draft-slot configuration and validation |
| 3     | Keeper configuration and validation     |
| 4     | Commissioner console                    |
| 5     | Team mobile interface                   |
| 6     | TV/display interface                    |

| Phase | Deliverable                                      |
|-------|--------------------------------------------------|
| 7     | Real-time synchronization and reconnect handling |
| 8     | Draft-night testing and hardening                |